

Euclid's Elements — Book I

Proposition 1

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

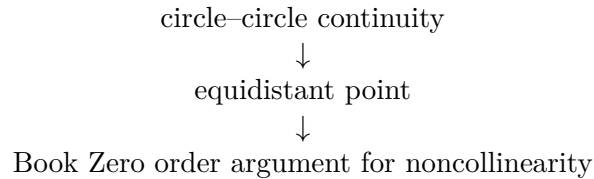
Constructing an equilateral triangle

1 Introduction

Proposition I.1 is the first construction in Euclid's *Elements*: given a nondegenerate segment AB , construct an equilateral triangle on that base. The familiar proof draws two circles of radius AB and uses a common point C as the third vertex.

The current formal reconstruction reveals more structure than this classical picture. The public theorem first obtains a point C equidistant from A and B , then proves separately that C cannot lie on the line AB . In the present production source, the equidistant-point theorem is no longer a provisional black box: it is proved from the project's general circle–circle continuity principle.

Thus I.1 already exhibits three layers:



This makes I.1 an especially clear test of the boundary between Euclid's visible construction and the lower Hilbert/Book Zero infrastructure.

2 Classical Configuration

Let AB be the given finite straight line, with $A \neq B$. Euclid draws the circle centered at A through B and the circle centered at B through A . If C is a common point of these circles, then

$$AC \cong AB, \quad BC \cong BA \cong AB.$$

Joining C to A and B therefore gives an equilateral triangle ABC .

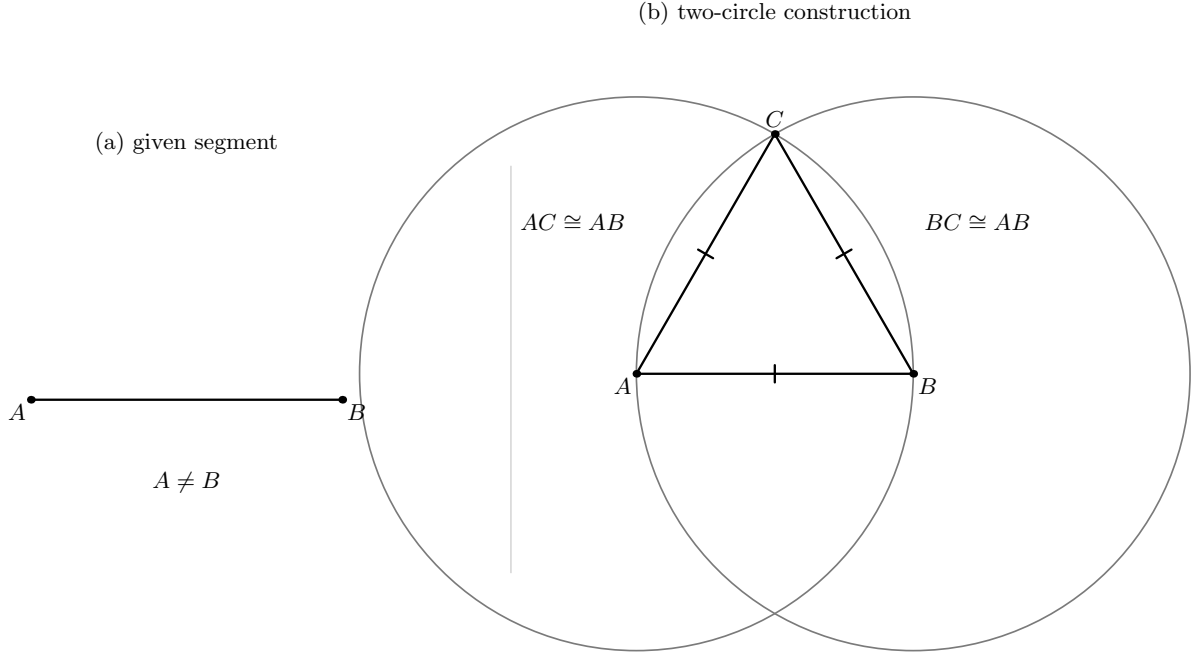


Figure 1: Euclid’s construction for Proposition I.1. (a) The input is a nondegenerate segment AB . (b) Two circles with centers A, B and radius congruent to AB meet at C ; the three sides of ABC are then congruent.

Foundational caution. The familiar two-circle picture suppresses the main existence issue. The permission to draw the two circles does not by itself prove that they meet. A rigorous axiomatization needs a circle-intersection or continuity principle. This is not a defect specific to the formal reconstruction: it is exactly the gap hidden by the classical diagram.

3 Statement

Theorem 1 (Euclid I.1). *For every nondegenerate segment AB there exists a point C such that ABC is a genuine equilateral triangle:*

$$\neg \text{Collinear}(A, B, C), \quad AC \cong AB, \quad BC \cong AB.$$

The current production declaration is:

```
theorem euclid_proposition_1
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B : Geo.Point)
  (hAB : A ≠ B) :
  ∃ C : Geo.Point,
    ¬ PrimCollinear Geo A B C ∧
    Geo.Congruent A C A B ∧
    Geo.Congruent B C A B := by
  ...
```

The formal statement makes the nondegeneracy of the triangle explicit. Euclid’s diagram visually places C off the base line; Lean requires this as a theorem.

Diagrammatic audit. The preserved Book1R chapter already reduced the classical construction from four incremental snapshots to two stable states. That reduction is sound and is retained. The formal part, however, was still represented mainly by a flowchart and described `hilbert_equidistant_point_exists` as a temporary existence principle. Both points are now outdated.

The audited chapter uses three geometric figures with seven panels:

2 classical panels + 2 continuity panels + 3 collinearity branches = 7 panels in 3 figures.

The three branch panels are required because the production proof performs a genuine betweenness trichotomy.

4 Classical Proof

Euclid's construction can be separated into the following steps.

1. By Postulate III, draw the circle centered at A through B .
2. Draw the circle centered at B through A .
3. Use a circle-circle intersection principle to obtain a common point C .
4. By Postulate I, join C to A and B .

Since C belongs to the first circle,

$$AC \cong AB.$$

Since C belongs to the second circle,

$$BC \cong BA.$$

Endpoint reversal does not change a segment, so $BA \cong AB$, and Common Notion 1 gives the required equality of the three sides.

The historical dependency structure is therefore

$$\text{Postulates I and III} + \text{circle-circle intersection} + \text{Common Notion 1} \longrightarrow \text{I.1.}$$

No earlier proposition of Book I is available or needed.

5 The Formal Existence Engine

The current source proves `hilbert_equidistant_point_exists`; it is not an axiom. The proof reduces the special equilateral configuration to the general continuity axiom `hilbert_circle_circle_intersection`.

Starting from $A \neq B$, the proof first extends AB beyond B . A transient point R supplies a ray from B . On that ray, Hilbert III.1 lays off a copy of BA , producing Y . Same-ray transport then yields

$$A - B - Y, \quad BY \cong BA.$$

The point R is used only to activate segment construction and disappears from the stable geometric state, so it is intentionally not drawn.

Now consider two circles represented synthetically by their centers and radius segments:

- circle 1 has center A and radius AB ;
- circle 2 has center B and radius BA .

The point A lies on circle 2 and is the center of circle 1. The point Y also lies on circle 2, because $BY \cong BA$. Since $A - B - Y$, the segment AB is a proper initial part of AY , so Y is outside circle 1 in the synthetic segment-order sense.

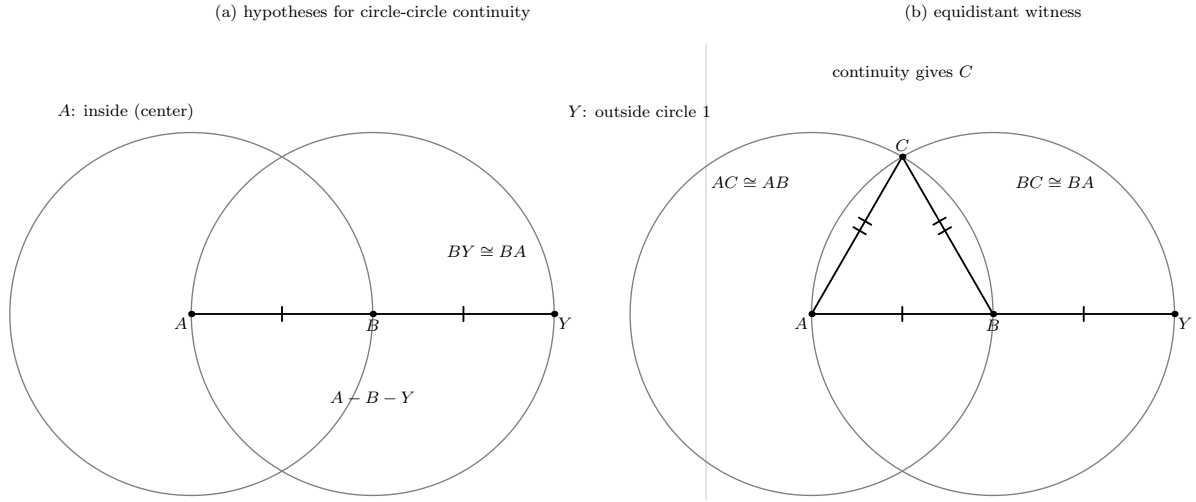


Figure 2: The formal continuity engine hidden below `hilbert_equidistant_point_exists`. (a) The construction gives $A - B - Y$ and $BY \cong BA$. Thus A is the center-case inside witness for circle 1, while Y lies on circle 2 and outside circle 1. The drawn circles are schematic: their visible crossing is not used as a proof. (b) The continuity axiom supplies a common point C , yielding $AC \cong AB$ and $BC \cong BA$.

The decisive source call is structurally:

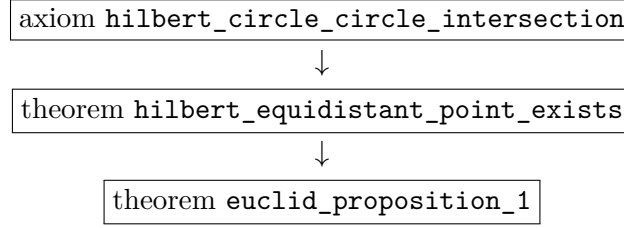
```
rcases
  hilbert_circle_circle_intersection
    Geo
    A B
    B A
    A Y
    ... with
  ⟨C, hAC_AB, hBC_BA⟩
```

After reversing the second radius orientation, the theorem returns exactly

$$AC \cong AB, \quad BC \cong AB.$$

6 Continuity Boundary

The source-level distinction is important. The chain is



The continuity axiom is stated synthetically. One circle contains two points X, Y on a second circle; X is inside the first circle (or is its center), and Y is outside it. The conclusion is the existence of a common point of the two circles. No coordinates, real-valued radii, or distance function are introduced.

For I.1 the generic axiom is specialized in a particularly economical way: $X = A$ is the center case, and Y is the explicitly constructed outside point. Thus the equidistant-point theorem is a genuine derived theorem from the project's accepted circle-continuity layer.

7 The Synthetic Noncollinearity Stage

Once C is obtained, the production theorem still has to prove that A, B, C form a genuine triangle. First it excludes $C = A$ and $C = B$. Both degenerate cases would make a nonzero segment congruent to a null segment; `bookZero_nullSegment1` rules this out.

Assume next, for contradiction, that A, B, C are collinear. Since the three points are pairwise distinct, Hilbert betweenness trichotomy gives exactly three cases:

$$A - B - C, \quad B - A - C, \quad A - C - B.$$

Each case contradicts the fact that a proper part cannot be congruent to the whole.

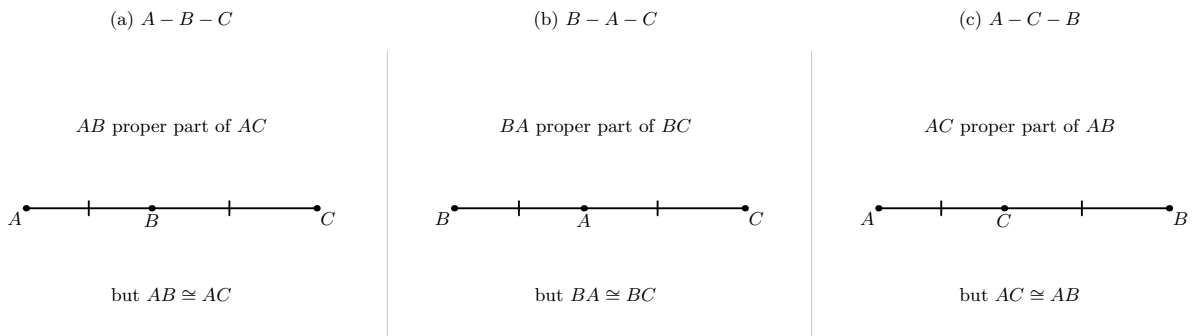


Figure 3: The three genuine branches of the formal collinearity reductio. In each order, one of the equidistant segments becomes a proper part of a segment to which it is congruent. Book Zero #45, `bookZero_45_partNotEqualWhole`, excludes all three cases.

More explicitly:

- If $A - B - C$, then AB is a proper part of AC , contradicting $AB \cong AC$.
- If $B - A - C$, then BA is a proper part of BC , while congruence transport gives $BA \cong BC$.

- If $A - C - B$, then AC is a proper part of AB , contradicting $AC \cong AB$.

Hence A, B, C are noncollinear.

8 Lean Formalization

At the Book I level the proof has three visible phases.

First, obtain the equidistant witness:

```
obtain ⟨C, hAC, hBC⟩ :=
  hilbert_equidistant_point_exists Geo A B hAB
```

Second, exclude the degenerate endpoint identifications using `bookZero_nullSegment1`. Third, assume collinearity and split the three possible orders:

```
rcases
  hilbert_between_trichotomy
    Geo A B C
    hAB hBCne hACne hCol with
  hABC | hBAC | hACB
```

Every branch is closed by `bookZero_45_partNotEqualWhole`, with only congruence orientation changes between the three cases.

This is a useful example of why theorem length alone is a poor guide to the geometric architecture. The public proof contains a substantial order argument, while its existence theorem contains a separate continuity construction beneath the interface.

9 Hilbert and Book Zero Components Used

9.1 Circle-circle continuity

The foundational existence input is `hilbert_circle_circle_intersection`. It is an explicit continuity axiom in the current interface. Proposition I.1 does not introduce a new local existence axiom.

9.2 Segment construction and order

Hilbert III.1 constructs the outside witness Y on a chosen ray. The order layer then turns same-ray information into the strict betweenness configuration $A - B - Y$, from which $AB < AY$ is represented synthetically.

9.3 Null-segment exclusion

The public theorem directly uses `bookZero_nullSegment1` to exclude $C = A$ and $C = B$. This declaration remains an active Book Zero/interface convenience axiom rather than one of Hilbert's historical congruence axioms. It should therefore be distinguished from the deliberately accepted continuity principle above.

9.4 Proper part versus whole

`bookZero_45_partNotEqualWhole` is the common contradiction engine for all three collinear orders. It makes explicit the elementary geometric fact that Euclid’s final diagram leaves visually obvious.

10 Relationship with the Foundations

I.1 has no earlier Book I theorem to call, so the foundational organization is exposed immediately. The classical and formal proofs now meet at a surprisingly close point: both ultimately require a circle-intersection principle. The difference is that the formal development states that principle explicitly and supplies the precise inside/outside witnesses needed to invoke it.

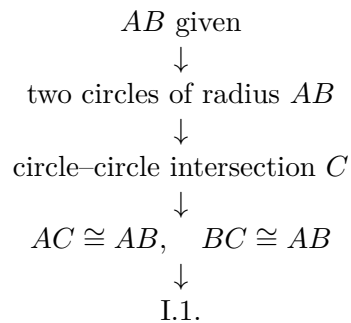
The remaining noncollinearity burden is then discharged without circles. This separation is conceptually useful:

continuity controls existence, order + congruence control nondegeneracy.

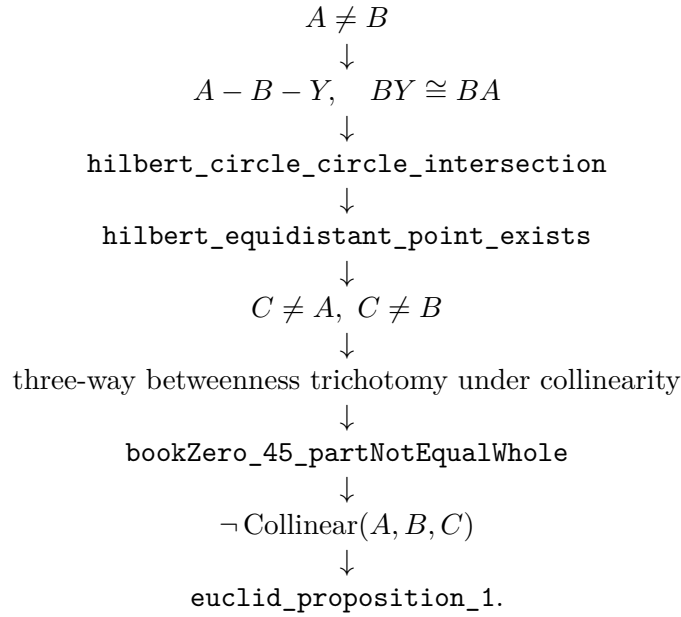
It is this second stage that demonstrates the role of Book Zero as a proof language rather than merely a repository of constructions.

11 Dependency Summary

The classical architecture is



The current formal architecture is



At the axiom boundary relevant to this chapter, the two explicit interface primitives to record are the accepted circle–circle continuity axiom and the separate Book Zero null-segment convenience axiom used in the endpoint degeneracy checks. There is no proposition-local axiom in `Proposition01.lean`.

12 Conclusion

Proposition I.1 is not merely the first easy construction in Book I. Once its hidden assumptions are exposed, it already displays the architecture that recurs throughout the reconstruction.

Euclid presents the geometry as two circles and their intersection. The current Lean development makes the same existence issue explicit through a general circle–circle continuity axiom, derives the special equidistant point from it, and then proves noncollinearity by a separate Book Zero order argument. The resulting theorem is therefore both a reconstruction of Euclid’s construction and an early demonstration of how the project divides proof obligations between continuity, congruence, and order.